

# Generated code	# Canonical solution
1 <code>def ap_sum(a,n,d):</code>	1 <code>def ap_sum(a,n,d):</code>
2 <code>total = 0</code>	2 <code>total = (n * (2 * a + (n - 1) * d)) / 2</code>
3 <code>for i in range(n):</code>	3 <code>return total</code>
4 <code>total += a + i * d</code>	
5 <code>return total</code>	

(a) Missed Mathematical Optimizations from MBPP-335

# Generated code	# Canonical solution
1 <code>def greatest_common_divisor(a:int,b:int)->int:</code>	1 <code>while b:</code>
2 <code>while a != b:</code>	2 <code>a, b = b, a % b</code>
3 <code>if a > b:</code>	3 <code>return a</code>
4 <code>a -= b</code>	
5 <code>else:</code>	
6 <code>b -= a</code>	
7 <code>return a</code>	

(b) Suboptimal Conditional Logic from HumanEval-13